Nyx Haile

Founding Engineer

xyn@mit.edu | LinkedIn | GitHub | Open to relocation

Experience

Partial 2025 – Present

Cofounder & CEO *London*

- Co-founded Partial, a prediction market that turns any relational analysis between assets into a tradable position.
- Ran customer discovery with self-funded traders, crypto executives, quantitative researchers and hedge fund managers.

Weiss Laboratory & The SHED, MIT June 2025 – Sept 2025

Undergraduate Researcher (UROP), generative AI for vascularised liver tissue *Cambridge, MA*

- Built the fluidics-design AI agent and wrote the master prompt for a proof-of-concept multi-agent LLM workflow that generates complex vascular STL designs from plain-text specifications; co-inventor on the MIT invention disclosure.
- Wrote a parametric vascular-network generator specification for the Cellvivo bioreactor, covering stochastic branching, tapering, R-tree collision avoidance and a 0.99 diffusion-coverage loop.

Vana Dec 2022 – July 2023

AI Developer *Remote*

- Engineered and iterated prompt templates for personalised AI-generated sticker portraits, tuning style and expression cues for identity retention and consistency.
- Helped organise the $25K Vana x MIT Generative AI Hackathon.

Selected Projects

Repositories marked private are available on request.

Strand *(private)* | Rust, FoundationDB *Capability-authenticated issue tracker*

- Built a federable, capability-authenticated issue tracker in Rust on FoundationDB, with line-level blame and content-addressed rationale.
- Shipped a 16-crate, ~71k-line Rust workspace through v1.0.0 GA (10/10 verification gates) and v1.1.0, then a v2.0 governed-execution prerelease.

Forge *(private)* | Rust, systemd, RPC *Local-first agent control plane*

- Built a Rust daemon, operator CLI, Claude Code hooks and provider proxy that coordinate coding-agent work across multiple git repositories.
- Added bounded, jittered exponential backoff to scheduler dispatch, seeded per task so retry waves de-correlate in production and replay exactly in tests.

howtowin.lol | Python, PyTorch, CUDA, Redis *Sequence model for post-game analysis*

- Designed a Dreamer-lineage RSSM sequence model for League of Legends combining static context, player priors and event sequences with outcome, next-event, next-frame and next-decision heads.

MIT Pokerbots | Python *Heads-up poker-variant bot, January 2026*

- Built a Python bot for MIT Pokerbots' heads-up poker variant; the team placed about 15th of 70.
- Implemented Monte Carlo equity estimation, with exact enumeration for discards, using quick and deep sampling paths to fit a per-decision compute budget.

Sherlock | Rust, Dolt *Token-usage forensics for agent CLIs*

- Built a Rust CLI that ingests agent-CLI logs (Claude Code, Codex, Copilot), attributes token usage per session, and reports what burns tokens without adding value.

Education

Massachusetts Institute of Technology May 2026

B.S. Mathematics, Minor in Economics *Cambridge, MA*

- Coursework: algorithms, theory of computation, computer systems security, cryptography, statistics.

Technical Skills

Languages: Rust, Python, TypeScript, SQL **ML:** PyTorch, NumPy, CUDA, GPU training and retrieval **Systems:** Linux/systemd, Docker, FoundationDB, PostgreSQL, SQLite, Redis **Web:** SvelteKit, React, Next.js